Design and Development of a Robotic Arm for a Soft Granular Jamming Gripper

Author : Kimberly Shanika Olanma Nweze

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Supervisor : Dr Majid Taghavi

Abstract

This project aims to design and develop a functional, lightweight 3 Degree of Freedom robotic arm for integration with a granular jamming gripper to assist in strawberry packaging. The arm is essential for the gripper's functionality. An articulated design was chosen after exploring rigid and soft structures. Initial concepts using wooden dowel as links were replaced by parallel rectangular beams based on a structural performance analysis. The first prototype, 3D printed at 70% infill, was not functional due to assembly challenges. A revised final version, printed at 40% infill and weighting 0.350 kg, featured motorised shoulder and elbow joints and resolved earlier design errors. Testing showed a rotational range of 295°, reach between 110mm-255mm and high precision (mean error: 3.95mm, Standard deviation: 1.9mm). The arm successfully supported the gripper's weight of 70g but exhibited a measured end-effector compliance of 1.55mm/g, leaving room for improvement. Aluminium was selected for the final fabrication due to its durability, machinability, and hight strength-to-weight ratio, using the lost PLA method. In conclusion, although the project's aim was achieved, further design improvements are required before transitioning to aluminium and integrating into real-world agricultural packaging applications.

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List of symbols

Symbol	Description	Unit	Found in section
W_{arm}	Maximum weight the motor supports	kg	3.1 Design Calculations
T_{safe}	Safe torque	Nm	3.1 Design Calculations
g	Gravity	N/kg	3.1 Design Calculations
d	Distance from the motor axis to the arm's centre of mass	m	3.1 Design Calculations
S	Section modulus	mm^3	3.1 Design Calculations
r	radius	mm	3.1 Design Calculations
I	Moment of inertia	$kg.m^2$	3.1 Design Calculations
b	Width of the beam	mm	3.1 Design Calculations
h	Height of the beam	mm	3.1 Design Calculations
I_{total}	Total moment of inertia	$kg.m^2$	3.1 Design Calculations
y_{max}	Maximum distance from the neutral axis	m	3.1 Design Calculations
π	Pi	/	3.1 Design Calculations

1. Introduction

Strawberries are among the most traded fruits globally [1]. Demand for soft fruits such as strawberries is projected to grow as the global population increases, necessitating an expansion in the cultivation and processing of these fruits [1]. In the soft fruit industry, processing relies heavily on human workers sorting, handling and packaging the fruit to prevent damage [1]. However, despite these efforts, manual processing still causes damage (e.g. bruising), reducing the shelf life of products, negatively affecting the farmers' economic benefits, and hindering the expansion of the strawberry industry [1].

The dependence on human labour introduces several challenges. Firstly, human labour is unable to keep up with the increasing demand for strawberries [1]. Secondly, the high labour requirement of processing leads to high costs for farmers [1]. Finally, the repetitive nature of the task poses a potential health risk, including musculoskeletal disorders in workers, and is detrimental to the sustainable growth of the strawberry industry [1].

A granular jamming gripper can be used to address these challenges. The gripper combines precision handling with intelligent control, making it suitable for managing delicate fruit such as strawberries. The integration of the gripper in the packaging process will reduce dependence on manual labour and increase productivity while maintaining the quality standards necessary for industrial packaging.

This report focuses on the design and development of the robotic arm attached to the granular jamming gripper. A robotic arm is a specialised robot made up of links and joints that allow rotational and translational motion. Its end-effector, located at the end of the arm, performs the primary task. The aim is to create a functional lightweight robotic arm with 3-Degrees of Freedom (DoF), striking an optimal balance between simplicity, cost-efficiency and sufficient range of motion. The arm plays a critical role in allowing the gripper to accurately pick-and-place delicate fruit from various positions within a range of environments, e.g. different container sizes and packing layouts. Without the arm, the functionality of the gripper will be significantly limited. Furthermore, the integration of a robotic arm enhances the gripper’s efficiency and scalability, making it better suited to meet the increasing demand for strawberries. Modularity and adaptability were also prioritised in the design to support future upgrades and broader integration across agricultural systems. The arm also serves as a platform for additional features such as object detection algorithms, sensor integration, and a modular control system, acting as a crucial link between the gripper and a wider robotic system.

The remainder of this report is structured as follows. Section 2 provides a literature review of various existing robotic arms, highlighting their limitations before focusing on those with objectives similar to this project. Section 3 describes the design process of the robotic arm, including a description of the final design and material selection. Section 4 outlines the testing procedures used to evaluate the arm’s performance and presents the results. Section 5 discusses the results, justifies robotic arm design, evaluates environmental and societal impacts, and presents a risk analysis. Finally, Section 6 lists the key conclusions, highlights encountered challenges and suggests directions for future improvements.

2. Literature Review

2.1. Modern Robotic Arms

Robotic arms have a wide variety of classifications, ranging from rigid structures with multiple DoF to soft continuum structures inspired by nature. Among rigid classifications, the most common structures include Articulated, Selective Compliance Assembly Robot Arms (SCARA), Polar, Cylindrical and Cartesian. Articulate robotic arms, which utilise revolute joints to mimic the flexibility of human arms, are widely used due to their precision and wide range of motion [2]. For example, K Kruthika et al. [2] present a 5-DoF articulated arm designed to assist elderly users with feeding tasks. In their work, inverse kinematics computes joint positions to achieve precise end-effector placement. The structure, machined from lightweight yet durable aluminium, is actuated by DC and stepper motors to deliver smooth, human-like movements under user control. Similarly, this project implements a 3-DoF articulated arm to achieve precision and flexibility in a packaging environment.

Articulated Robotic arms are also known for their ability to handle diverse materials [3]. R Jain et al. [3] utilise SolidWorks to virtually model and analyse an articulated manipulator without constructing a physical prototype, reducing development costs. Their finite-element analysis compares structural steel and aluminium alloy 356 under various loading scenarios, concluding that structural steel is preferable for its superior stiffness and lower deformation under end-effector forces [3].

On the other hand, SCARA robots combine high speed, repeatability, and simplified maintenance, making them ideal for pick-and-place applications [4]. In the study by K Phasale et al. [4], a 4-DoF SCARA system (comprising one linear and three rotational joints) was fabricated from acrylic to leverage its transparency and low mass. DC servo and geared motors were integrated to drive each link, achieving rapid, accurate positioning in controlled environments and validating the configuration's suitability for industrial assembly lines.

In contrast to rigid arms, soft continuum robots derive their compliance and motion capabilities from elastic deformation, drawing inspiration from biological appendages such as elephant trunks, octopus arms, and snakes [5]. These backbone-less manipulators wrap around various objects to achieve secure grasps and adaptive manipulation. H Yang et al. [6] introduce a silicone-based soft arm actuated by shape-memory alloy coils. By embedding miniature Hall sensors, the system measures bending and orientation in real time, enabling a proportional–integral–derivative (PID) controller with compensation algorithms to achieve multidirectional motion with reliable accuracy.

M W Hannan et al. [7] developed a continuum arm resembling an elephant's trunk, consisting of four serially connected segments of varying diameter, each driven by a hybrid cable and spring servo system. Their kinematic model informs motion planning across the entire manipulator, and experiments demonstrate successful grasping of objects, confirming the efficacy of cable-spring actuation in continuum architectures.

Overall, rigid and soft robotic arms exhibit strengths. Articulated and SCARA arms excel in precision, speed and material handling, while continuum arms excel at adaptability and gripping of irregular objects. These advances in design, modelling and actuation continue to expand the applications of robotic arms across multiple industries.

2.2. Agricultural Robotic Arms and Criticism

In recent years, agricultural practices have been significantly influenced by advancements in robotic systems, especially in harvesting, sorting and packaging soft fruit.

Recent advancements in agricultural robotics showcase diverse approaches to automated fruit harvesting. The Agrobot E-Series employs 24 stainless-steel, military-grade arms guided by LiDAR for obstacle detection and RGB cameras for fruit recognition, achieving high-precision picking [8]. Similarly, the Vegebot uses a 6-DoF arm with a custom gripper and region-based convolutional neural network (R-CNN) vision to classify and extract soft produce at 82% accuracy, though its 31.7s cycle time indicates potential for optimisation [8]. Another example is the Noronn AS Strawberry Harvester, which utilises a 5-DoF arm and R-CNN detection, reaching a 74.1% harvest success rate, highlighting the challenges of unstructured field conditions [8].

Other prototypes adopt specialised sensing and end-effector designs. For instance, a 5-DoF arm with a pneumatic gripper and binocular servo vision succeeded in hydroponic settings by handling occluded fruit, though its performance degrades under outdoor variability [1].

Diverse methodologies are used for the integration of robotic arms in agriculture. Mechanical cut-and-separate systems integrate machine vision with cutting tools on mobile platforms to automate harvesting in controlled environments, reducing labour dependency [1]. Another approach involved Hybrid Harvest Assist Robots, which improve manual picking by transporting trays and improving ergonomics, thus mitigating musculoskeletal strain and enhancing efficiency [1]. In addition, fully autonomous systems combine advanced perception and robotic arms to potentially automate entire soft fruit harvesting [1].

Despite progress, key limitations prevent scalability. Soft fruits often cluster, causing misidentification and missed picks, while ambient light fluctuations impair vision algorithms [8]. Additionally, fragile produce is often damaged due to grippers' issue with balancing firmness and gentleness [8]. Finally, long cycle times decrease efficiency, and high development costs confine these technologies to prototypes [8].

Addressing these gaps requires advanced solutions. This may include the use of dual-arm systems and advanced obstacle-separation algorithms for clustered fruit, depth cameras for 3D localisation, and standardised LED lighting to stabilise vision [8]. Furthermore, adaptive soft grippers with tuneable compliance could minimise damage, and refined motion-planning in dual-arm systems may reduce cycle times [8].

In conclusion, achieving scalability and cost-efficiency in agricultural robotics relies on a focus on mass production and modularity during the development stage, ensuring a resilient yet affordable systems [8]. Although limitations remain, emerging developments continue to improve precision, efficiency and feasibility of existing systems, making the way for a more automated and sustainable agricultural future.

2.3. Narrowing the Scope

Despite the advanced features of state-of-the-art robotic arms, their complexity and scale make them unsuitable for this project's objectives. Most are relatively large and feature more than 5-DoF, whereas this project aims for a lightweight, 3-DoF robotic arm. Valuable insights for this project were provided by M Farman et al. [9, who designed a 3-DoF arm with a colour-sorting mechanism for picking and placing lightweight objects. The authors' work highlights the significance of appropriate link shapes for efficiency while stressing the role of proper design and simulations in achieving proposed objectives.

Similarly, A K Saini et al. [10] designed a 3-DoF robotic arm for pick-and-place tasks, emphasising the benefit of kinematic analysis and practical design principles. The arm was designed using Fusion 360 and 3D-printed using PLA (Polylactic Acid) filament, balancing lightweight components with structural integrity. For forward and inverse kinematics, the Denavit-Hartenberg (DH) convention was employed, ensuring accurate joint angle calculations for precise end-effector positioning. Each joint was modelled to mimic human arm movements, enhancing natural motion and precision.

Although both studies contributed to the design process of the robotic arm, they present recognised limitations. By relying on standard servo motors, M Farman et al. [9] and A K Saini et al. [10] limited their arms 's ability to handle complex, heavier loads. Additionally, A K Saini et al. [10] use of PLA constrains the arm's durability and strength, further hindering its performance with heavier loads or for prolonged use. While modular, the design by A K Saini et al. [10] lacks scalability for industrial applications.

These challenges highlight opportunities to improve motor capabilities and material selection, balancing cost-effectiveness with functional robustness for broader applications.

3. Design Process

3.1. Design Calculations

The aim of the design process was to create a functional robotic arm with 3 revolute joints, which enable controlled rotation along a single axis [11]. Providing a 360° range, the base joint allows fruit to be picked from any direction, while the shoulder and elbow joint extend the reach horizontally. Only two XC430-W240 motors were available for the project, due to budget constraints. The base and elbow joints were motorised, while the shoulder joint was manually adjustable using a bolt to preserve the 3-DoF. The base joint is essential for a 360° rotation, and the elbow joint connects to the gripper, which is crucial for handling delicate fruit.

Servo motors were selected for their ability to precisely control angular position, velocity, and shaft acceleration [12]. Despite this advantage, they have torque limitations, with a stall torque of 1.9 Nm [13]. Torque refers to the rotational force produced by the motor, measured in Newton-meters (Nm), whereas stall torque is the maximum torque a motor can provide momentarily [13].

To prevent overheating or damage, the safe torque (T_{safe}) was set at 80% of the stall torque, resulting in a value of 1.52 Nm. Using this torque, the maximum weight the motor could support for the candidate axis distance of 200mm was calculated using Equation 1:

$$W_{arm} = \frac{T_{safe}}{g \times d} = \frac{1.52\ Nm}{9.81 \times 0.2(m)} \approx 0.775kg \quad (1)$$

In this equation, g is gravity, and d represents the distance from the motor axis to the arm's centre of mass.

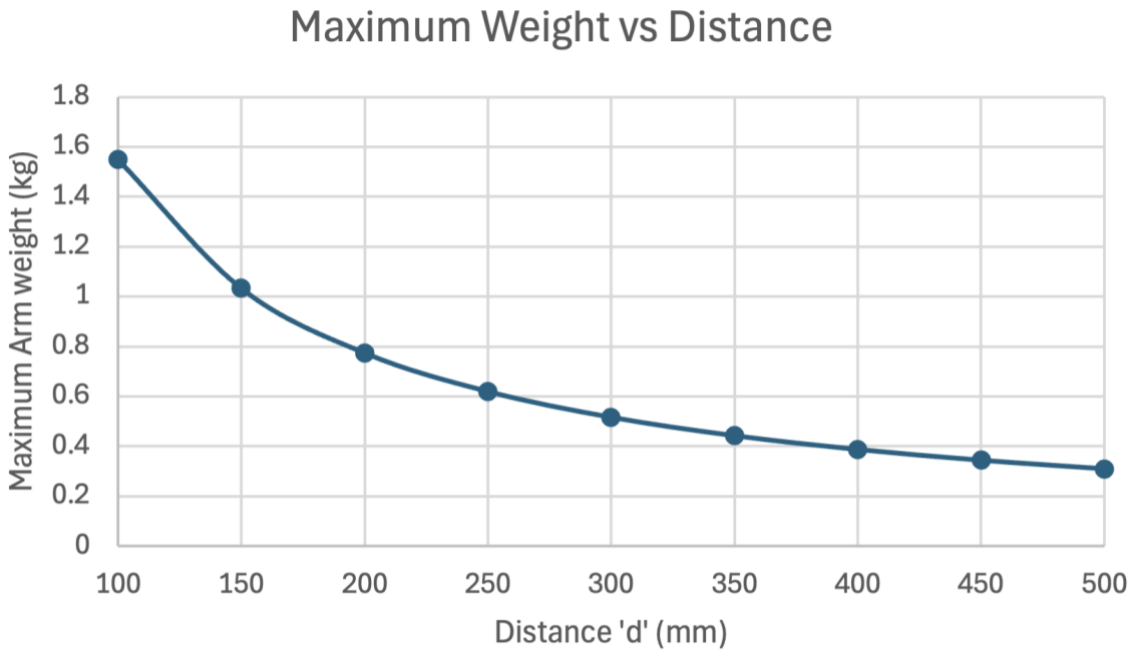


Figure 1: Correlation between d and the maximum combined arm and load weight

Figure 1 illustrates an inverse relationship between d and the maximum load the motor can sustain. A shorter d allows the motor to support a heavier load. Balancing reach and weight, the design aimed d at 200 mm, enabling a maximum arm weight of 0.775 kg as shown in Equation 1. Accordingly, Link 1 and Link 2 were set to 200 mm and 100 mm, respectively, providing a total reach of 300 mm while remaining within the motor's load-bearing capacity.

During the design process, the structural efficiency of different link types was investigated. Initially, wooden dowels (solid cylindrical beams) were used. However, studies indicated that these have a low strength-to-weight ratio [9]. To improve the design, MATLAB code was developed to compare the section modulus of solid cylindrical beams with that of two parallel rectangular beams. The section modulus (S) is a geometric property indicating a beam's resistance to bending and depends on the cross-sectional shape and dimensions [9].

The code first defined 100 data points and evaluated cross-sectional areas ranging from 0 to 7000 mm². For the cylindrical beam, where r is the radius, S was derived from the cross-sectional area using Equation 2:

$$S = \frac{\pi r^3}{4} \tag{2}$$

For each rectangular beam, where b is the width and h is the height of the beam. The moment of inertia was calculated using Equation 3:

$$I = \frac{bh^3}{12} \tag{3}$$

The moment of inertia of the connector was also considered, accounting for its area and distance from the neutral axis. The total moment of inertia (I_{total}) was then divided by the maximum distance from the neutral axis to compute S, as shown in Equation 4:

$$S = \frac{I_{total}}{y_{max}} \tag{4}$$

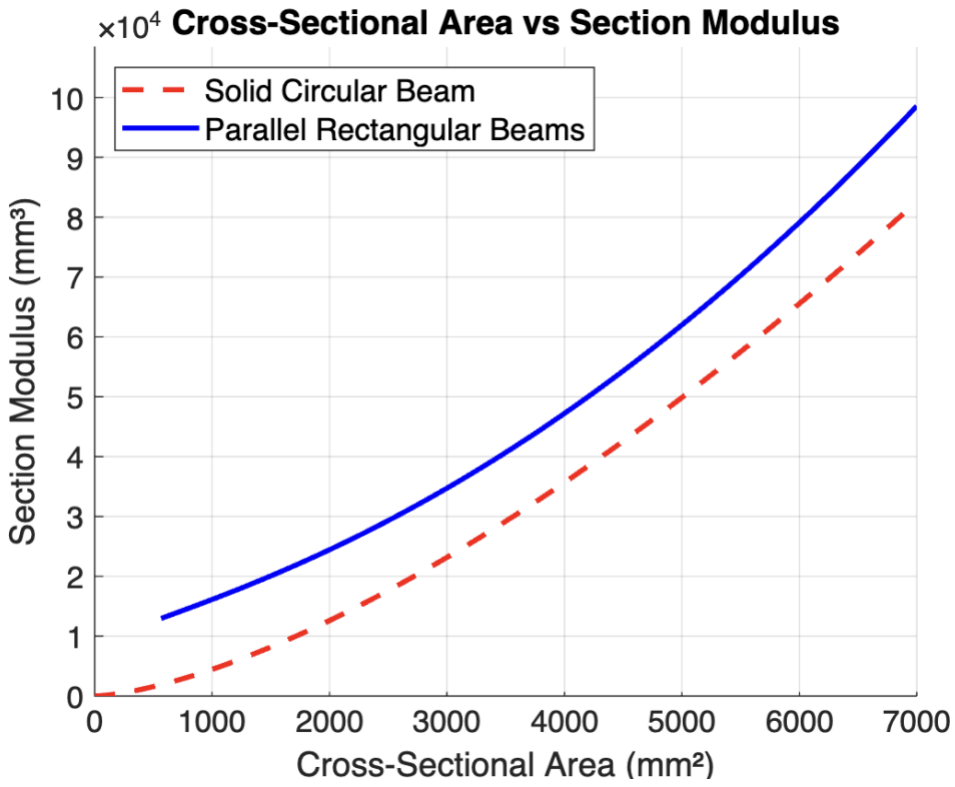


Figure 2: Cross-Sectional Area against Section Modulus

Figure 2 illustrates that across all cross-sectional areas, the parallel rectangular beams consistently exhibited a higher section modulus, indicating superior strength and resistance to bending. Based on these findings, the design was updated to incorporate parallel rectangular beams for overall improved structural performance.

3.2. First Prototype

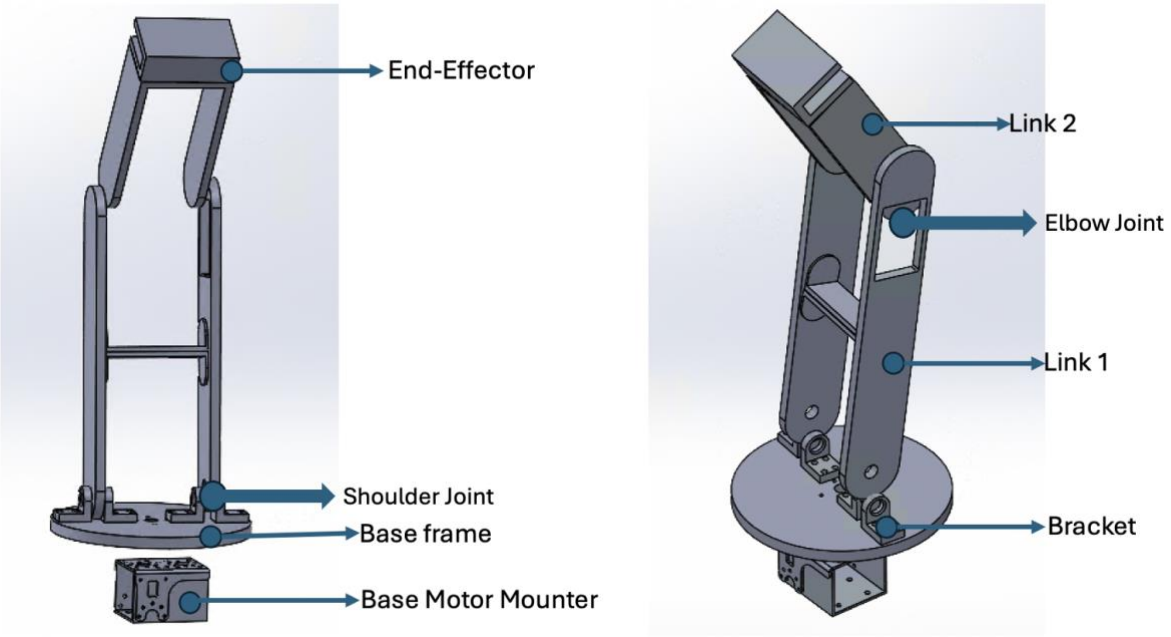


Figure 3: CAD Assembly of First Prototype

SolidWorks was used to design and simulate the robotic arm. The first prototype of the arm was 3D printed using PLA at 70% infill to minimise weight. Figure 3 illustrates the first prototype assembly. The base motor is mounted vertically on the base motor mounter, allowing 360° rotation.

The manually adjustable shoulder joint uses brackets screwed into the base frame, connected to Link 1 through a hex bolt for versatility and strength. Link 1 consists of two rectangular beams joined by a bracket, with a cutout to accommodate the servo motor rotating Link 2. This motor controls the elbow joint, which enables arm extension.

Link 2 connects to the end effector, which holds the granular jamming gripper, keeping it perpendicular for precise fruit handling.

3.2.1. Design Considerations

It is essential to consider various factors when designing a product. The high cost of robotic arms limits implementation in agriculture, as farmers question their economic benefits [1]. The modular, simple design of this robotic arm enables scalability and easier integration into packaging lines, making it more accessible.

If commercialised, it would reduce labour needs, lower costs, and improve operational efficiency. Robots also work continuously, reducing downtime losses caused by human rest periods. This automation helps reduce health risks from repetitive tasks and, by increasing efficiency, may lessen dependence on harmful pesticides, supporting more sustainable farming practices [1]. Overall, the design promotes both economic and environmental benefits for the agricultural industry.

3.2.2. Challenges and Design Improvements

Several challenges were encountered during the assembly of the first prototype, prompting multiple design improvements.

The base motor mounter failed to support the motor as intended. To address this, the mounter was redesigned so that the motor sits inside it rather than on top. This allowed for secure attachment from the sides, ensuring stability and ease of assembly.

The base frame also presented several challenges. The rotating face of the motor is extruded, a detail that was not considered initially. Integrating the two parts required the use of long screws, which was impractical. Furthermore, the screw holes intended for bracket attachment were misaligned and too close together, preventing integration with Link 1. The frame itself was overly thick and heavy, adding unnecessary strain on the motor. These problems were resolved by redesigning the base into a rounded rectangular shape, removing excess material to reduce weight. The part that connects to the motor was reshaped to perfectly match the motor face, eliminating the need for long screws. The thickness was reduced from 8 mm to 5 mm, and infill was lowered from 70% to 40% to further reduce weight.

The misaligned holes were corrected by redesigning the brackets. The original L-shaped brackets covered excessive space on the base frame, limiting screw placement. This was resolved by creating simplified brackets that allowed screws to attach directly into them from the base frame.

Link 1, consisting of two rectangular beams, also required improvements. The holes for the motor and bearing were misaligned. To address this, the bearing hole was repositioned slightly to ensure alignment. Additionally, the motor cut-out was too tight, which was solved by slightly increasing the motor outline for a proper fit.

The connector between the two beams of Link 1 was initially designed as two parts that screw together to form an ‘I’ shape. They were misaligned, making it impossible to assemble correctly. This was resolved by redesigning it as a single ‘I’ shape piece that screws directly into both beams.

Insufficient space was left for the bearing between Links 1 and 2, and the design of Link 2 did not accommodate the extruded motor face. This was solved by shortening the Link 2 and modifying the motor interface to perfectly fit.

3.3. Final Prototype

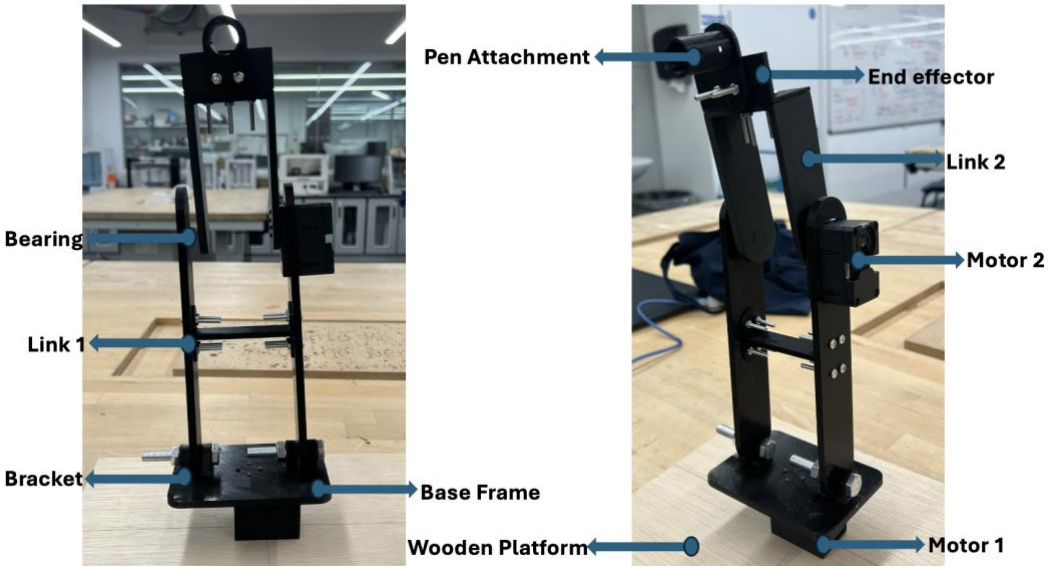


Figure 4: Assembly of Final Prototype

Figure 4 illustrates the final prototype, redesigned in SolidWorks based on improvements from the initial version. All previous integration challenges were addressed, allowing for smooth physical assembly. Modularity was a key design focus, with components designed for easy tuning and replacement during the fabrication process to enhance longevity and repairability thus increasing the robot's long term economic benefit.

The end effector can be screwed into either the granular jamming gripper or the pen attachment, which is designed to hold various pen sizes for testing purposes. Additionally, a new component was developed to attach to Link 2, enabling the integration of a Raspberry Pi High Quality Camera. This supports the potential implementation of object detection algorithms, further enhancing the system's functionality.

This prototype combines practical and educational objectives with industrial design principles, highlighting functionality and durability while maintaining simplicity in assembly and maintenance.

3.3.1. Material Selection and Fabrication Process

Material selection is critical for the performance of the robotic arm. For the final prototype, parts were 3D printed using PLA filament with 40% infill to reduce weight. Lightweight materials are crucial for effective weight distribution and motor efficiency, contributing to precision and reliability in robotic operation [10]. The weight of the final prototype is 0.350kg.

While PLA is suitable for prototyping, it has limitations in strength compared to metals. The final design aims to use aluminium, a material with a high strength-to-weight ratio [12]. Aluminium is capable of withstanding bending loads due to its superior section modulus, is easy to machine while being resistant to corrosion [12]. These properties make it ideal for creating a durable, lightweight robotic arm capable of handling delicate tasks.

The transition to aluminium can be achieved using the lost PLA method, where PLA parts are either placed in sand moulds or encased in plaster and melted to create a cavity for molten aluminium [14]. This improves the arm's reliability and longevity, overcoming PLA's limitations while maintaining precision.

4. Testing and Results

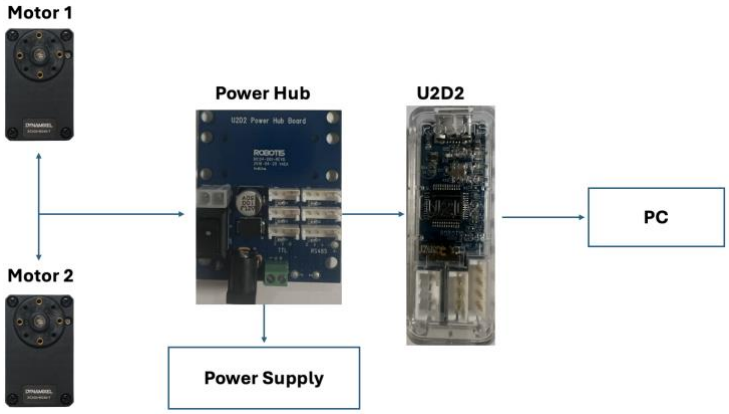


Figure 5: Testing Configuration Diagram

During testing, motor 1 (base) and motor 2 (Link 2) were connected to a power hub, which was linked to a power source. The hub was also connected to the U2D2, a compact USB communication converter used to control DYNAMIXEL servo motors using a PC [15].

4.1. Workspace Test

The arm was controlled to draw circles using the pen end effector to determine its maximum and minimum limits, as well as the rotational angles of the motors. This data was then used to define the workspace.

4.1.1. Results

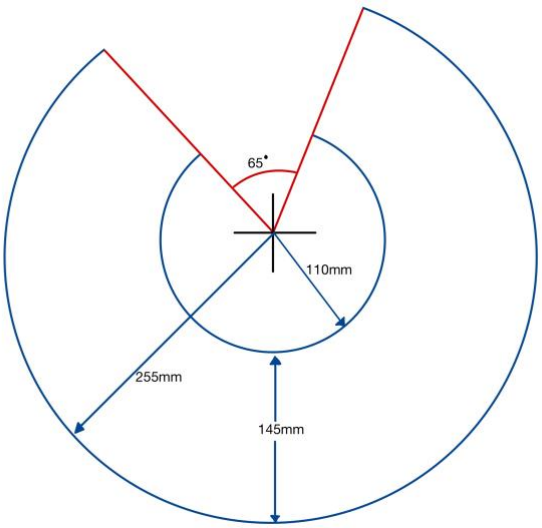


Figure 6: 2D Workspace of Robotic Arm

Figure 6 illustrates the 2D workspace of the robotic arm. The arm's maximum and minimum reach were measured at 255 mm and 110 mm, respectively. It failed to achieve full 360° rotation, missing 65°, as measured with a protractor.

4.2. Precision and Accuracy Test

The arm was repeatedly controlled to the same point to measure precision and accuracy. This was repeated 20 times, and the difference between each point and the initial point was measured. Ideally, there should be no difference between the points. The initial point was located 220mm from the centre, within the arm's workspace.

4.2.1. Results

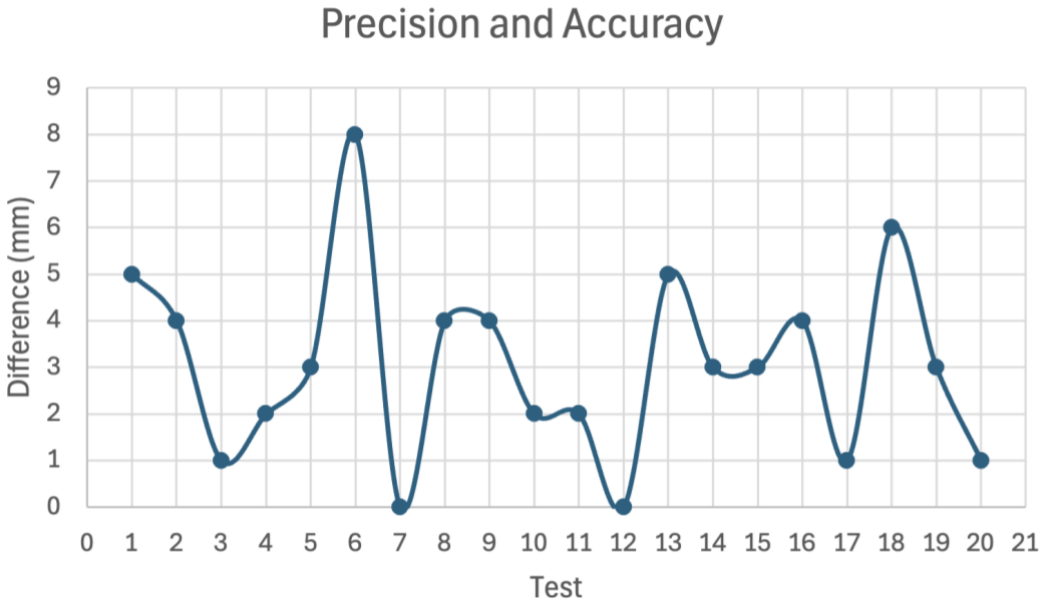


Figure 7: Difference between the initial point and the measured point

Figure 7 illustrates the variation between the initial point and each measured point across all tests. The mean difference was calculated as 3.95mm, with a standard deviation of 1.99mm, indicating the arm's consistency.

4.3. Weight Capacity Test

The previous test procedure was repeated while incrementally increasing the end effector's weight by 10 g each time, for a total of eight repetitions. The starting weight was 22 g, representing the pen end effector. The granular jamming gripper weighs approximately 70g, making it the test's target weight. This test aimed to determine the arm's capability of lifting the gripper, with the initial position maintained at 220 mm.

4.3.1. Results

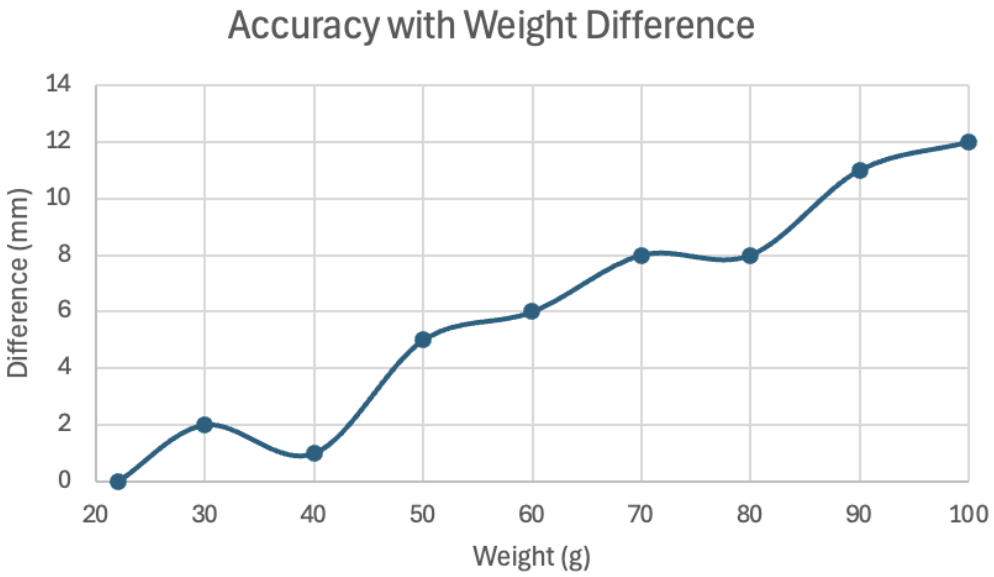


Figure 8: Difference between the initial point and the measured point with different weights

Figure 8 illustrates a positive correlation between weight and the measured difference from the initial point. As the weight increased, the difference also rose, with several points falling outside the expected range. The test concluded at 100 g, as the primary goal was to confirm the arm's ability to lift 70 g, which it successfully achieved. The mean and standard deviation of the measured differences were calculated as 6.63 mm and 3.37 mm, respectively. The average increase in difference for each gram added was calculated to be 0.155 mm/g.

5. Discussion

5.1. Test Analysis and Design Implications

5.1.1. Workspace Test

During testing, it was observed that the rectangular shape and size of the wooden platform restricted inward movement, suggesting the arm's minimum reach could be shorter than measured. Reducing the platform size and adopting a circular shape in future designs could help expand the workspace. The reduced maximum reach from the predicted 300mm is due to the links not being connected end-to-end; rather, each link overlaps slightly at the joint. This overlap, not accounted for in the design, explains the 45 mm difference and can be corrected by increasing link lengths. The arm's inability to rotate 360° is due to software-imposed motor limits, not a hardware flaw, because manual 360° rotation is possible when the motors are off. The arms workspace can be increased with a better control system, allowing it to rotate 360°. Additionally, motors offer high resistance when powered, preventing manual joint movement. Noise levels monitored during this test indicated a smooth and quiet performance, with louder sounds only occurring when the arm exceeded its workspace before quickly returning to the start position.

5.1.2. Precision and Accuracy Test

The low standard deviation of 1.99 mm indicates good precision, and the mean difference of 3.95 mm further confirms the arm's accuracy. Given the gripper has a larger contact area than the pen tip, this level of error is well within acceptable limits. The maximum deviation recorded was 8 mm, which is minor and unlikely to affect performance. Overall, the results demonstrate that the arm can consistently and accurately reach its intended position. During this test, the noise level was moderate, with noticeable sound generated as the arm moved between positions, but it did not interfere with operation.

5.1.3. Weight Capacity Test

The arm weighs 0.350 kg, within the calculated maximum safe load of 0.775kg. The bottom motor rotates without challenge, confirming the system can reliably support the granular jamming gripper and multiple strawberries. This test revealed that from 80 g upwards, the pen touched the paper prematurely, likely due to the arm being weighed down. At 90g, a slight bounce was observed as the arm lifted, and at 100 g, it touched the surface with noticeably greater force. These effects can be mitigated with a more refined control system or by reprinting Link 2 and the end effector with a stronger filament such as PETG (Polyethylene Terephthalate Glycol) for added rigidity. This issue is expected to be resolved when switching to aluminium due to its superior structural integrity.

Additionally, while this test recorded a higher mean and standard deviation, the smaller dataset prevents meaningful comparison with the previous test. The calculated average increase in difference of 0.155mm/g while small in isolation, accumulates rapidly in packaging applications with varying loads and could potentially impact positional accuracy. This value reflects the system's mechanical compliance under varying loads and reinforces the need for a stronger material.

Finally, noise levels increased with heavier loads, likely due to the motors working harder to compensate for the added weight, making this test the loudest overall.

5.2. Justifying Rigid Design over Continuum Alternatives

For this project, a rigid robotic arm structure was selected due to its superior compatibility with end-effectors, particularly grippers. Since one of the project's objectives is integration with a soft granular jamming gripper, a rigid design was the more appropriate choice. Continuum arms, by contrast, do not typically require end-effectors, as they use the entire structure to manipulate objects [5]. However, this characteristic does not align with the project's goal of incorporating a specific gripper type. Additionally, while continuum arms excel in navigating environments with obstacles [5], soft fruit packaging is predictable and repetitive. These conditions are compatible with rigid arms, which offer the precision and repeatability needed for such environments [3]. Therefore, considering both the integration requirements and the nature of the task, a rigid arm design was better aligned with the project's overall objectives.

5.3. Environmental and Social Impact

The arm has several environmental impacts. Its precision reduces strawberry bruising, extending shelf life and lowering food waste, which comprises 6-8% of annual greenhouse gas emissions [16]. Additionally, the use of aluminium enhances durability, lowering the frequency of replacements. Although aluminium production has a high carbon footprint, it is highly recyclable [17], enabling parts to be reused without compromising material integrity. However, improper recycling of aluminium parts will contribute to landfill accumulation. Additionally, the incorporation of the arm adds to the energy usage of the production line, especially with a suboptimal control system.

Socially, the arm would reduce repetitive strain injuries inflicted on workers [1]. Furthermore, the arm would mitigate the effects of labour shortages, especially in agriculture, where seasonal work is variable.

However, widespread implementation of robotic arms in fruit packaging may create job displacement. This would be partially mitigated by the requirement of workers to operate the robotic arm, which requires additional training of the workforce. Overall, the robotic arm may simultaneously contribute to a reduction in demand for manual labour and higher training costs for remaining workers.

5.4. Risk Analysis

When integrated into a packaging environment, an aluminium robotic arm poses several mechanical risks. Continuous, repetitive motion over time may lead to the deterioration of joints, especially at the base and the elbow joint. Although aluminium offers structural rigidity, under sustained loading, the arm could experience minor deformation, affecting precision. Loose or incorrect connections of components could also lead to the arm malfunctioning, performing inaccurately during pick-and-place tasks.

Unintended collision between the arm links, nearby objects and workers could occur due to limited mechanical constraints or overextension. Additionally, vibration during rapid movements may increase long term strain on components and reduce reliability. Finally, environmental contaminants such as dust or food particles from the packaging line could build up at joints, leading to malfunctions. Though aluminium is resistant to corrosion, performance may still be affected by sustained exposure.

These risks could be mitigated through regular maintenance to prevent malfunctions and provide potential reinforcements to critical areas. Also, proximity sensors could be incorporated to prevent collision with workers and other objects. Finally, joint limits could be set to prevent the arm from exceeding its lifting limits.

6. Conclusion

A functional and lightweight 3-DoF robotic arm compatible with a granular jamming gripper for strawberries pick-and-place task was successfully developed. A rigid structure was selected over a continuum structure due to its greater suitability for precise, repetitive tasks typical in packaging environments. Designed in SolidWorks and 3D printed using PLA at 40% infill, the final prototype weighed 0.350kg and featured a motorised base and elbow joint. The final version of the arm is intended to be fabricated in aluminium using the lost PLA method for longevity and improved durability.

Exploring the various design possibilities made it difficult to define the initial concept. However, iterative troubleshooting helped refine the structure, particularly by resolving early issues with structural integrity. Testing showed a maximum reach of 255 mm, failing to reach the predicted 300 mm due to link overlap, which can be addressed by marginally extending the links. It also indicated that the minimum reach is potentially smaller than recorded, the arm was restricted by the size and shape of the wooden platform. This limitation could be addressed by replacing the wooden platform with a more compact, circular design. The arm exhibited high precision, with a mean error of 3.95 mm and a standard deviation of 1.99 mm, and reliably supported weights of the 70 g gripper and additional load. However, increasing the load on the end effector revealed that beyond 80g, premature contact and increased bounce occurred. This reflects the measured end effector compliance of 0.155mm/g, highlighting the importance of a stiffer material (e.g. aluminium) or advanced control compensation to preserve accuracy under varying loads.

While PLA was appropriate for early prototyping, its structural limitations constrained the arm's performance. Exploring stronger alternatives, such as PETG, could have enhanced performance. Additionally, although the tests provided a good insight into the arm’s accuracy and strength, a wider range of test and more diverse loading scenarios would increase the reliability of the results.

Environmentally, the arm shows potential in reducing food waste and the recyclability of aluminium mitigates the greenhouse gas emissions. Socially, it may help address labour shortages and reduce repetitive strain injuries. However, disadvantages include increased energy consumption and potential job displacement.

Future work will focus on further refining the design to address mechanical risks such as component degradation and unintended collisions. These upgrades are essential for transitioning the prototype into a robust industrial solution, particularly through joint reinforcement and sensor integration.

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Appendix

Test 1 – Missing Arc length calculations

Arc length = $\frac{\theta}{360°} \times 2\pi r$

$\theta = 65° \; r = 255mm$

Arc length = $\frac{\theta}{360°} \times 2\pi 255 = 289.3mm$

Test 2 – Table of results and Calculations

Test	Measured Distance	Difference
1	225	5
2	224	4
3	221	1
4	222	2
5	223	3
6	228	8
7	220	0
8	224	4
9	224	4
10	222	2
11	222	2
12	220	0
13	225	5
14	223	3
15	223	3
16	224	4
17	221	1
18	226	6
19	223	3
20	221	1

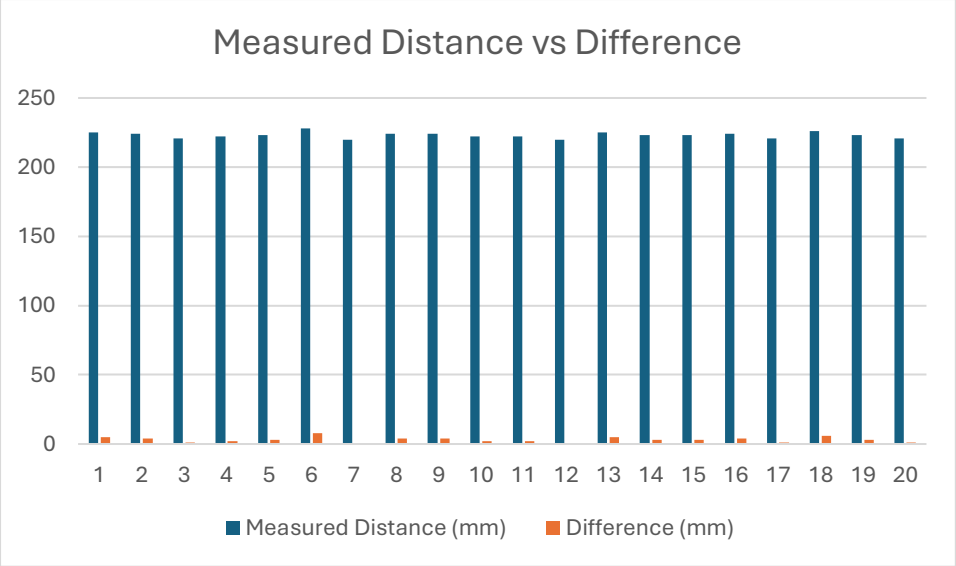


Figure A1: Comparison between the measured distance and the difference from the initial point to help emphasis how small the difference between each test is.

Sum of difference = 61mm

Mean of difference $=\frac{61}{20} = 3.05mm$

Difference	Difference - mean	$(\text{Difference} - \text{mean})^2$
5	1.95	3.8025
4	0.95	0.9025
1	-2.05	4.2025
2	-1.05	1.1025
3	-0.05	0.0025
8	4.95	24.5025
0	-3.05	9.3025
4	0.95	0.9025
4	0.95	0.9025
2	-1.05	1.1025
2	-1.05	1.1025
0	-3.05	9.3025
5	1.95	3.8025
3	-0.05	0.0025
3	-0.05	0.0025
4	0.95	0.9025
1	-2.05	4.2025
6	2.95	8.7025
3	-0.05	0.0025
1	-2.05	4.2025

Standard deviation = $\sqrt{\frac{\text{Sum of (Difference} - \text{mean)}^2}{20}} = \sqrt{\frac{78.95\ \text{mm}}{20}} = 1.99$ mm

Test 3 – Table of results and Calculations

Test	Weight (g)	Measured Distance (mm)	Difference (mm)
1	30	222	2
2	40	221	1
3	50	225	5
4	60	226	6
5	70	228	8
6	80	228	8
7	90	231	11
8	100	232	12

Sum of difference = 53mm

Mean of difference =$\frac{53}{8} = 6.625mm$

Difference (mm)	**Difference - mean**	$(\textbf{Difference} - \textbf{mean})^2$
2	-4.625	21.390625
1	-5.625	31.640625
5	-1.625	2.640625
6	-0.625	0.390625
8	1.375	1.890625
8	1.375	1.890625
11	4.375	19.140625
12	5.375	28.890625

Standard deviation = $\sqrt{\frac{\text{Sum of (Difference} - \text{mean)}^2}{8}} = \sqrt{\frac{107.875\ \text{mm}}{8}} = 3.62$ mm

Average increase in difference measured per gram added calculations

Slope of regression line = $\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$

Total Weight (g)	Added weight from 22g (g)	Difference
22	0	0
30	8	2
40	18	1
50	28	5
60	38	6
70	48	8
80	58	8
90	68	11
100	78	12

Means:

$$\bar{x} = \frac{0+8+18+28+38+48+58+68+78}{9} = 38$$

$$\bar{y} = \frac{0+2+1+5+6+8+8+11+12}{9} = 6$$

$\boldsymbol{x}$ $\boldsymbol{(g)}$	$\boldsymbol{y}$ $\boldsymbol{(mm)}$	$x-\bar{x}$	$y-\bar{y}$	$\boldsymbol{product}$
0	0	-38	-6	228
8	2	-30	-4	120
18	1	-20	-5	100
28	5	-10	-1	10
38	6	0	0	0
48	8	10	2	20
58	8	20	2	40
68	11	30	5	150
78	12	40	6	240

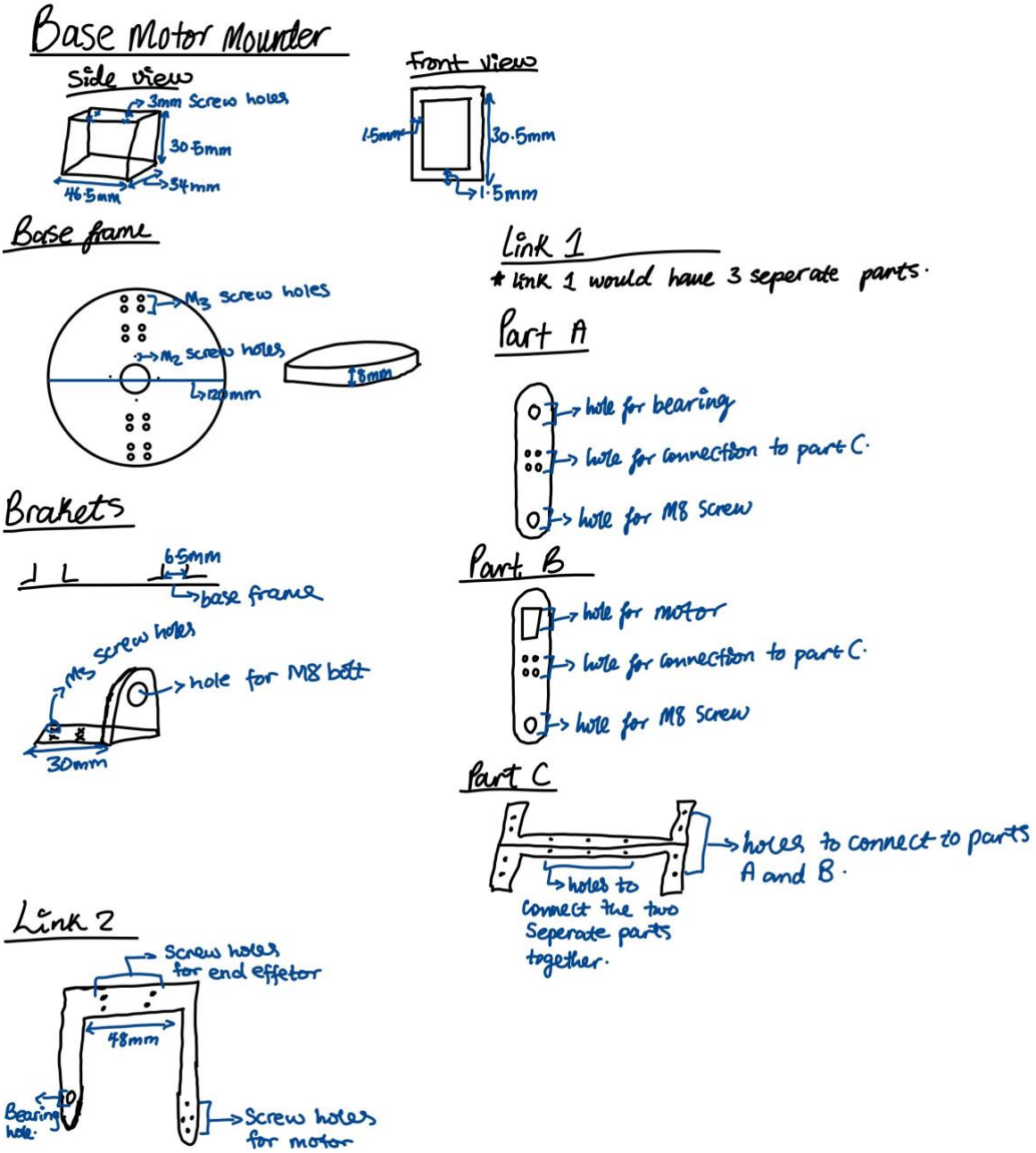
Numerator = sum of all the product = 908

$x - \overline{x}$	$(x - \overline{x})^2$
-38	1444
-30	900
-20	400
-10	100
0	0
10	100
20	400
30	900
40	1600

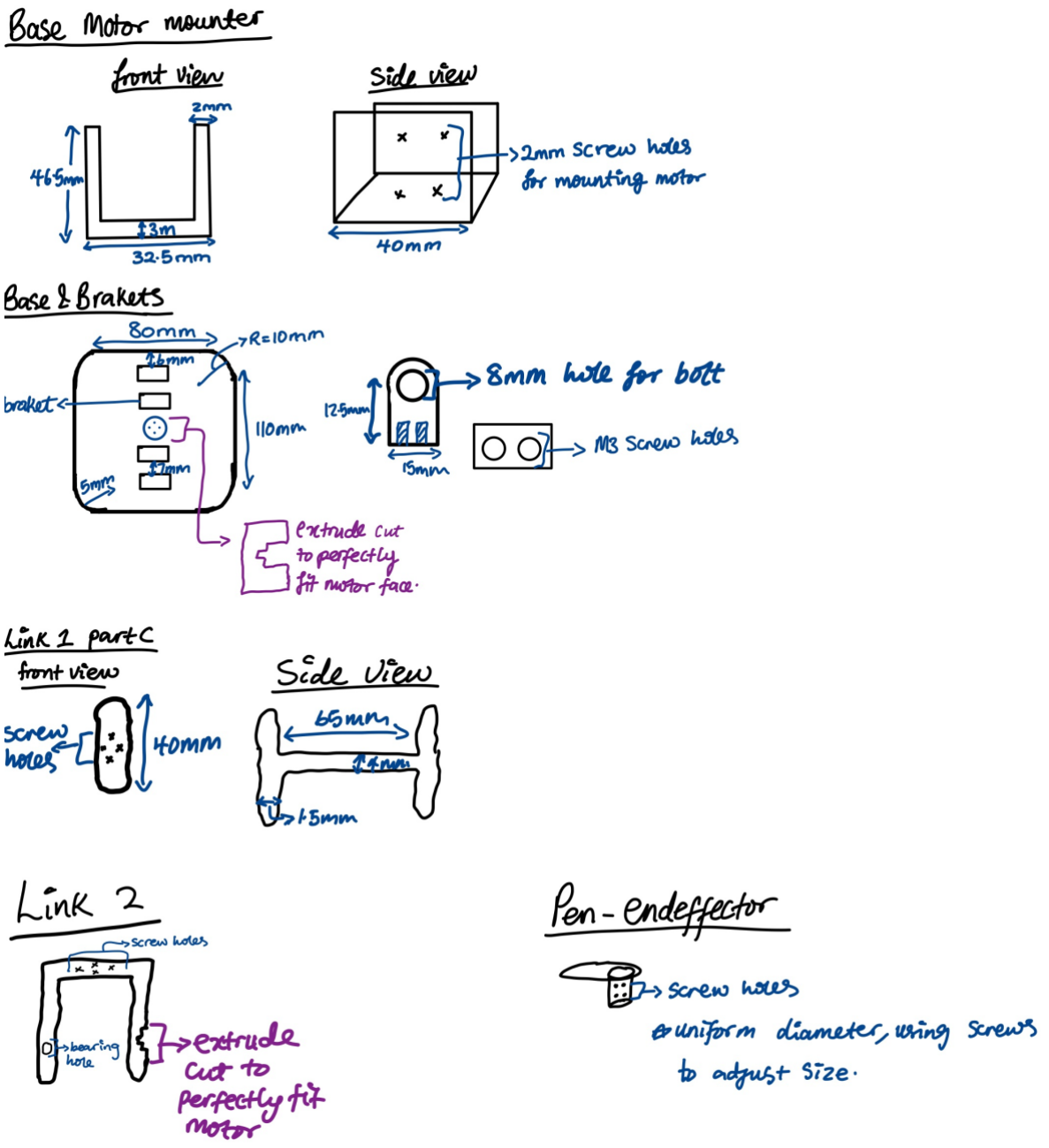
Denominator= sum of the squared numbers = 5844

$$\therefore Average\ increase\ per\ gram = \frac{908}{5844} \approx 0.1553\ mm/g$$

Concept sketches for the first prototype arm



Concept sketches for the final prototype arm



Picture of joints pre-assembly



